**HUMPHREY VIRTUAL FARMS LLC // LEVEL-5 SOVEREIGN C2 ENCLAVE**

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CLASSIFICATION: SOVEREIGN LEVEL-5 EXECUTIVE DELIVERABLE // PUBLIC/PRIVATE PARITY

## Project Ebony v8.0.0: Deterministic Zero-Cloud OT/SCADA Interlock Testbed & WORM Merkl

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DOD CDAO TRADEWINDS SOLUTIONS MARKETPLACE // v8.0.0 GROUND-UP RESUBMISSION DOSSIER

NEW TITLE: Project Ebony v8.0.0: Deterministic Zero-Cloud OT/SCADA Interlock Testbed & WORM Merkle Provenance Architecture (TRL 4 Assessor-Aligned Resubmission)

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**PART 1: NEW PORTAL ABSTRACT (COPY & PASTE INTO TRADEWINDS SUBMISSION FORM)**

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[OCT 2026 v8.0.0 RESUBMISSION DELTA: Explicitly resolves all prior 9-26-3703/10-26-3976 Assessor Feedback with new 4-Pillar Rubric proof, TRL 4 software-vs-hardware target separation, & a 5-vendor AoA matrix.] 1. PROBLEM: Per GAO-21-81/GAO-24-106137, DoD’s 500+ installations & $4.2B+ OT energy systems rely on digital PLCs scanning at 50-200ms. Per CISA AA23-144A/AA24-038A (Volt Typhoon) & TRISIS, adversaries corrupt PLC logic while spoofing HMIs. Since high-voltage arc-flash occurs in <4.16ms & thermal excursions in 6.0ms, 50-200ms scans are 8x-33x too slow. 2. SOLUTION (TRL 4 TESTBED): Project Ebony v8.0.0 runs 100% offline on 127.0.0.1 loopback (0% cloud, 636 C2 modules, 85/85 unit tests passed, 17/17 Red-Team vectors blocked, 110-control NIST 800-171/800-82 mapping). Brain 1 models <50us safety loops, 2.04us Modbus FC05 trips & a 36.75ns SiC MOSFET cutoff target; Brain 2 runs local NDE/PCI diagnostics (0 actuator wire); Brain 3 seals a 52-table SQLite SHA-256 WORM DAG. 3. STATE OF THE ART (AoA): Outclasses Siemens S7-1500/Rockwell (50-200ms, 150-450W), Triconex SIS (10-50ms), passive IDS Dragos/Nozomi (0 cutoff authority, 300-1200W), & Jev Cloud AI (>250ms RTT) via a modeled 36.75ns cutoff (1,360,544x faster than 50ms PLCs) at 28.5W on 24V DC. 4. TRANSITION: Unbundled 26-CLIN Console (CLIN 0001 Pilot $2,450/mo; 1-Node Bundle $895K/yr) & $1.485M 12-Mo Phase-1 Prototype OTA (10 USC 4022, $150K M1A) -> 4022(f) production (100% DFARS 252.227-7018 IP).

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**PART 2: 4-QUADRANT RESUBMISSION DELTA MATRIX (RESOLVING ALL PRIOR ASSESSOR FEEDBACK)**

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[DELTA-01] Prior Critique (Criterion 1): Quantify DoD OT scale, cite federal threat advisories, and prove why legacy PLC scan rates fail.

-> NEW IN v8.0.0: Anchored in GAO-21-81 & GAO-24-106137 (500+ installations, 300k+ buildings, $4.2B+ OT energy) and Joint CISA/NSA/DoD Advisories AA23-144A & AA24-038A (Volt Typhoon) plus TRISIS. Demonstrated on an animated physics timeline that <4.16ms arc-flash backfeeds and 6.0ms excursions outrun 50-200ms PLCs by 8x to 33x.

[DELTA-02] Prior Critique (Criterion 2): Separate TRL 4 software emulation from TRL 6/7 hardware claims and verify zero cloud dependency.

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-> NEW IN v8.0.0: Strictly separated our live TRL 4 local loopback software testbed (127.0.0.1, 0% cloud, 636 C2 modules, 85/85 passing unit test suites, 17/17 Red-Team blocks, 52 SQLite WORM tables, 1,420 Academy Qs) from our modeled Phase-1 OTA hardware fabrication targets (36.75ns SiC MOSFET cutoff, <50us Cortex-M7 loop, 28.5W SWaP).

[DELTA-03] Prior Critique (Criterion 3): Provide a quantitative Analysis of Alternatives (AoA) benchmarking Ebony against named competitors.
-> NEW IN v8.0.0: Added a dedicated 5-Vendor Quantitative AoA Matrix & SWaP Comparison Chart contrasting Project Ebony against Siemens S7-1500/Rockwell ControlLogix (50-200ms, 150-450W), Schneider Triconex SIS (10-50ms, 200-600W), Dragos/Nozomi passive IDS (0 physical cutoff authority, 300-1,200W), and Jev Cloud AI (>250ms API RTT).

[DELTA-04] Prior Critique (Criterion 4): Present an audit-proof commercialization roadmap, per-node pricing, and 10 U.S.C. 4022 OTA schedule.
-> NEW IN v8.0.0: Scrubbed all ambiguous deployment language; presented our Red-Team-certified client evaluation build, staged civil bridge NDE blueprints ($250/sortie drone model vs $4,600 snooper trucks), 26-Module CLIN Price Schedule (CLIN 0001-0026), and a 4-milestone $1.485M Phase-1 Prototype OTA Gantt schedule ($150K M1A Mobilization).

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**PART 3: 5-VENDOR QUANTITATIVE ANALYSIS OF ALTERNATIVES (COMPETITOR MATRIX)**

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* METRIC: Primary Architecture
- Project Ebony v8.0.0 (1AHA8) : Tri-Brain Software Testbed (TRL 4) + Modeled 36.75ns Analog Guillotine
- Siemens / Rockwell PLCs : Digital OS/RTOS PLC (5-20Hz Cyclic Scan)
- Dragos / Nozomi Guardian : Passive Network IDS / Cloud-Hybrid SIEM
- Schneider Triconex SIS : Triple-Modular Redundant Digital PLC
- Cloud AI Wrappers (Jev) : Cloud API Classification Wrapper
* METRIC: Cutoff Latency (Design Target vs Legacy)
- Project Ebony v8.0.0 (1AHA8) : 36.75 ns (Modeled SPICE Target) / 2.04 us (Emulated Modbus FC05)
- Siemens / Rockwell PLCs : 50,000,000 - 200,000,000 ns (50 - 200 ms)
- Dragos / Nozomi Guardian : NONE (0 Physical Cutoff Authority)
- Schneider Triconex SIS : 10,000,000 - 50,000,000 ns (10 - 50 ms)
- Cloud AI Wrappers (Jev) : 250,000,000+ ns (> 250 ms Cloud API RTT)
* METRIC: Speed & Physics Advantage
- Project Ebony v8.0.0 (1AHA8) : Modeled 1,360,544x Faster than 50ms PLC (Targets <4.16ms Arc-Flash)

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- Siemens / Rockwell PLCs : 8x - 33x Too Slow for <4.16ms Arc-Flash / 6ms Excursion
- Dragos / Nozomi Guardian : Requires Human Operator (> 250 ms Reaction)
- Schneider Triconex SIS : Vulnerable to TRISIS/Triton Firmware Injection
- Cloud AI Wrappers (Jev) : Total Failure Under EMCON / SATCOM Severance

* METRIC: Target SWaP Power Draw

- Project Ebony v8.0.0 (1AHA8) : 28.5 W Modeled Target (< 30W on 24V DC Tactical Bus)
- Siemens / Rockwell PLCs : 150 W - 450 W (Rack + HMI + Active Cooling)
- Dragos / Nozomi Guardian : 300 W - 1,200 W (Appliance Server + Cloud)
- Schneider Triconex SIS : 200 W - 600 W (TMR Chassis)
- Cloud AI Wrappers (Jev) : Data Center GPU Cluster Required

* METRIC: Cloud Dependency (Verified Live)

- Project Ebony v8.0.0 (1AHA8) : 0% (100% Air-Gapped Localhost 127.0.0.1 Software Testbed)
- Siemens / Rockwell PLCs : Proprietary Firmware / Windows Engineering Workstation
- Dragos / Nozomi Guardian : High (Cloud Threat Feeds / Linux OS)
- Schneider Triconex SIS : Proprietary Firmware (Compromised in TRISIS)
- Cloud AI Wrappers (Jev) : 100% Dependent on External Commercial Cloud

* METRIC: WORM Cryptographic Ledger (Verified Live)

- Project Ebony v8.0.0 (1AHA8) : SHA-256 WORM Merkle DAG (52 Verified SQLite Tables on Disk)
- Siemens / Rockwell PLCs : Unauthenticated Modbus/CIP by Default
- Dragos / Nozomi Guardian : Mutable Syslog / SIEM Export
- Schneider Triconex SIS : Standard Event Buffer (Non-Merkle)
- Cloud AI Wrappers (Jev) : Mutable Cloud Database Logs

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**PART 4: 5-SCENE MULTI-ARCHITECTURE VIDEO SCRIPT**
(EBONY_V8_CDAO_RUBRIC_VERIFICATION_20261028.mp4)

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>>> SCENE 1 (0:00 - 0:50) // NEW v8.0.0 RESUBMISSION DELTA MATRIX (4-QUADRANT BEFORE vs. AFTER)
RUBRIC: Section VII.e Mandatory Resubmission Gate: Point-by-Point Assessor Feedback Resolution
SCRIPT: Welcome to the October 2026 version 8.0.0 architectural verification of Project Ebony, presented by Jeffery Humphrey, Founder, President, and Chief Executive Officer of Humphrey Virtual Farms LLC, CAGE Code 1AHA8, UEI S1M4ENLHTDH5. We engineered this completely new verification briefing from the ground up to directly resolve every item of Assessor Feedback from our prior Tradewinds filings. On screen in our four-quadrant Resubmission Delta Matrix, you see our four structural upgrades: Quadrant One quantifies the DoD operational technology threat using GAO-21-81, GAO-24-106137, and

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CISA Volt Typhoon advisories. Quadrant Two establishes strict audit transparency--separating our live TRL 4 local software emulation testbed from our modeled Phase-1 hardware design targets. Quadrant Three introduces a five-vendor Quantitative Analysis of Alternatives against Siemens, Rockwell, Triconex, Dragos, Nozomi, and cloud AI. Quadrant Four details our 26-module CLIN price schedule and our $1.485 million Phase-1 Prototype OTA pathway.

**>>> SCENE 2 (0:50 - 1:40) // CRITERION 1: DEFINING THE PROBLEM & LOGARITHMIC PHYSICS TIMELINE**

RUBRIC: Criterion 1: Defining the Problem (GAO Citations, Volt Typhoon & <4.16ms Arc-Flash Physics)

SCRIPT: Pillar One: Defining the Problem. According to U.S. Government Accountability Office reports GAO-21-81 and GAO-24-106137, the Department of Defense operates more than 500 military installations, 300,000 facilities, and $4.2 billion in annual energy and microgrid assets controlled by commercial PLCs scanning at 50 to 200 milliseconds. Joint CISA, NSA, FBI, and DoD Advisories AA23-144A and AA24-038A confirm that state-sponsored Volt Typhoon actors pre-position inside critical energy, water, and logistics OT corridors. As shown on our physics timeline below, high-voltage arc-flash backfeeds destroy equipment in 4.16 milliseconds, and microreactor thermal excursions occur in 6.0 milliseconds. Legacy 50-to-200 millisecond digital PLC scan loops arrive 8 to 33 times after physical destruction has already occurred.

**>>> SCENE 3 (1:40 - 2:40) // CRITERION 2: ACCELERATING THE MISSION (TRI-BRAIN SCHEMATIC & LIVE TESTBED)**

RUBRIC: Criterion 2: Accelerating the Mission (Live 127.0.0.1 Testbed, 85/85 Tests & 17/17 Red-Team Blocks)

SCRIPT: Pillar Two: Accelerating the Mission. On screen is our Tri-Brain architectural schematic alongside live execution of our TRL 4 software emulation testbed on air-gapped localhost 127.0.0.1, operating with zero percent commercial cloud dependency and zero external APIs. Brain One executes deterministic safety boundary logic in software while our SPICE mathematical model targets a 50-microsecond bare-metal loop, 2.04-microsecond RS-485 Modbus FC05 trip, and a 36.75-nanosecond silicon-carbide MOSFET cutoff for Phase-1 hardware fabrication. Brain Two executes offline structural, pavement, and telemetry anomaly detection with zero physical or logical authority to override Brain One safety interlocks. Brain Three chains every frame into an append-only SHA-256 WORM Merkle Directed Acyclic Graph across 52 verified SQLite tables, proven live across 636 C2 modules, 85 of 85 passing unit test suites, and 17 of 17 blocked Red-Team cyber vectors.

>>> SCENE 4 (2:40 - 3:40) // CRITERION 3: ADVANCING STATE OF THE ART (5-VENDOR AoA & SWaP CHART)

RUBRIC: Criterion 3: Quantitative Analysis of Alternatives vs. Siemens, Rockwell,

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Triconex, Dragos & Cloud AI
SCRIPT: Pillar Three: Advancing the State of the Art. Our Quantitative Analysis of Alternatives table and SWaP bar chart benchmark Project Ebony directly against four classes of industry incumbents. First, digital PLCs like Siemens S7-1500 and Rockwell ControlLogix require 50 to 200 milliseconds per scan and draw 150 to 450 watts. Second, triple-modular redundant safety systems like Schneider Electric Triconex require 10 to 50 milliseconds, draw 200 to 600 watts, and were compromised by TRISIS firmware injection. Third, passive OT network monitors like Dragos Platform and Nozomi Guardian draw up to 1,200 watts and possess zero physical cutoff authority, requiring over 250 milliseconds for human intervention. Fourth, commercial cloud AI wrappers like Jev require 250-millisecond cloud round-trips and fail completely under EMCON. Project Ebony combines a verified zero-cloud WORM enclave with a modeled 36.75-nanosecond cutoff target--1.36 million times faster than a 50-millisecond PLC--at a modeled 28.5 watts on a 24-volt DC battery bus.

**>>> SCENE 5 (3:40 - 4:35) // CRITERION 4: 26-MODULE CLIN SCHEDULE & $1.485M PHASE-1 OTA GANTT**
RUBRIC: Criterion 4: Audit-Proof Commercialization Readiness, Unbundled CLINs & 10 **U.S.C. 4022 OTA**
SCRIPT: Pillar Four: Business Model and Acquisition Pathway. Every commercial and state artifact in our codebase is transparently engineered as audit-proof deployment readiness: a standalone Red-Team-certified client evaluation package, staged civil bridge and pavement diagnostic blueprints modeling a cost reduction from $4,600 snooper-truck inspections to $250 per autonomous drone sortie, and a complete 110-control NIST 800-171 and 800-82 architectural mapping with formal C3PAO and SPRS certification scheduled in Phase 1. Government buyers can procure our unbundled 26-module console starting at CLIN 0001 Base Pilot for $2,450 per month or CLIN 1004-A Full 1-Node Bundle for $895,000 per year, or award our 12-month, $1.485 million Fixed-Price Prototype OTA under 10 U.S.C. Section 4022, beginning with a $150,000 Day 15-to-30 mobilization tranche to fabricate physical Cortex-M7 and MOSFET boards and advance from TRL 4 to TRL 6, while Humphrey Virtual Farms LLC retains 100 percent DFARS 252.227-7018 Background IP. Thank you.